

Code No: 156CW**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year II Semester Examinations, August/September - 2021****SOFTWARE TESTING METHODOLOGIES****(Common to CSE, IT)****Time: 3 Hours****Max. Marks: 75**

Answer any five questions
All questions carry equal marks

- 1.a) State and explain various dichotomies in software testing.
b) Explain the life cycle of Bug. [8+7]
- 2.a) What are control and sequence bugs? How can they be caught?
b) Explain heuristics procedures for sensitizing paths. [7+8]
- 3.a) Compare data flow and path flow testing strategies.
b) Explain transaction-flow graph implementation with example. [7+8]
- 4.a) What is program slicing? Explain Dynamic program slicing.
b) Explain predicates of domain testing with examples. [8+7]
- 5.a) Write the role of path expression and path predicates in testing.
b) Write the procedure to count the minimum number of paths in a graph. [7+8]
6. Demonstrate through truth tables the validity of the following theorems of Boolean algebra:
a) Associative Laws
b) De morgan's theorems for three variables
c) Distributive Law. [5+5+5]
- 7.a) Explain about good state and bad state graphs. How to handle bad state graphs.
b) Write short notes on:
i) Transition Bugs
ii) Dead States. [7+8]
- 8.a) Explain the features of the Jmeter Testing environment.
b) How to record tests and set checkpoints in win runners? Explain. [7+8]

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ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) Define path testing. [2]
- b) What is the difference between an error and a bug? [3]
- c) Write a short note on random testing. [2]
- d) What is the significance of data flow testing? [3]
- e) Write a short note on path expressions. [2]
- f) List out the different operators that are used to solve any boolean algebra. [3]
- g) Define a transition bug. [2]
- h) What is good state graph? [3]
- i) Define a connection matrix. [2]
- j) List the applications of graph matrices. [3]

PART – B**(50 Marks)**

- 2.a) Differentiate between testing and debugging.
 - b) Describe the model for testing. [5+5]
- OR**
3. Classify the different types of bugs and explain. [10]
- 4.a) State and explain the transaction flow testing techniques.
 - b) Compare static slicing with dynamic slicing. [6+4]
- OR**
5. How developers and testers treat nice and ugly domains? Illustrate with the help of examples. [10]
- 6.a) Illustrate maximum path count arithmetic with an example.
 - b) Describe the usage of regular expression in flow anomaly detection. [6+4]
- OR**
- 7.a) Justify the use of decision table implementation for designing test cases.
 - b) Explain the procedure for specification validation using KV charts. [5+5]

8. Explain the following terms:
a) Design guideline for building finite state machine
b) Inessential finite state behavior. [5+5]

OR

- 9.a) Write short notes on testability tips.
b) Summarize the concept transition testing. [5+5]

- 10.a) Describe node-term reduction optimization.
b) Give a brief summary on relations. [5+5]

OR

11. Write an algorithm for node reduction using matrix operations and explain. [10]

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Answer any five questions
All questions carry equal marks

- 1.a) Discuss about implementation and application of path testing.
- b) Describe instrumentation and sensitization in transaction flow testing. [8+7]
- 2.a) Explain about approximate number of paths with suitable example.
- b) Discuss in detail about state bugs. [8+7]
- 3.a) Describe software implementation of state graphs.
- b) Explain about partition algorithm with example. [8+7]
- 4.a) Briefly explain about consequences of bugs.
- b) Discuss about three and four variable KV charts with examples. [8+7]
- 5.a) What are dataflow anomalies? Explain about dataflow anomaly state graph.
- b) Write and explain testability tips of state testing. [8+7]
- 6.a) Distinguish the following: i) Modularity vs efficiency ii) Function vs structure
- b) Describe motivational overview of graph matrices. [8+7]
- 7.a) Explain basic concepts of path products and path expressions.
- b) Discuss about node-reduction algorithm. [7+8]
8. Explain the following:
 - a) Testing one-dimensional domains
 - b) Coding bugs
 - c) Identity elements. [5+5+5]

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